

From Generative to Embodied Intelligence: Network Evolution for Distributed Large-AI-Model Agents

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Abstract—Large AI models are evolving from generators that return digital content, to agents that execute multi-step workflows, and further to embodied agents that perceive, decide, and act in the physical world. This evolution changes the volume, temporal structure, reliability target, and failure cost of network traffic. A generative request tolerates elastic delivery if response quality is preserved; an agentic workflow accumulates failure probability over tool calls; and an embodied loop can become unsafe when multimodal observations or action tokens are late, asynchronous, or distorted. This survey develops a network-centric comparative framework across these three stages. It shows that the same base network quantities—latency, rate, reliability, traffic density, and connection density—remain relevant but increase by orders of magnitude, while model-side emphasis moves from content fidelity, through end-to-end task success, to closed-loop control and safety. It then traces architecture from cloud-centric inference to stateful edge orchestration and hierarchical, partially decentralized multi-agent control. Enabling technologies are mapped to a token plane, a transport plane, and a coordination plane. A token-recognition case study translates observable traffic and endpoint-declared metadata into deployable service classes without requiring payload inspection. Finally, the paper identifies tokenizer compatibility and incrementally deployable token-aware protocols as prerequisites for networks that connect distributed intelligence rather than merely transport bits.

Index Terms—Agentic AI, embodied AI, edge intelligence, large AI models, multi-agent systems, networked control, semantic communications, token communications, ultra-reliable low-latency communications.

I. INTRODUCTION

LARGE AI models are moving beyond a single prompt-response exchange. Generative models synthesize text, images, audio, and video; agentic models decompose goals, invoke tools, retrieve memory, and cooperate with other models; embodied models close the loop between multimodal sensing and physical actuation. We use *large AI model* (LAM) as an umbrella term for these foundation-model-based systems, including large language models (LLMs), multimodal large language models (MLLMs), world models, and vision-language-action (VLA) models. The three stages are not mutually exclusive model families. Rather, they represent increasingly stringent interaction contracts between intelligence, a network, and an environment.

Mobile and edge networks are therefore changing roles. In the generative stage, a network primarily transports prompts, context, and output tokens to a remote accelerator. In the

agentic stage, it also binds tools, model context, persistent memory, and multiple expert models into a reliable transaction. In the embodied stage, the network becomes part of the plant: it carries camera, depth, point-cloud, inertial, joint-state, and action streams whose age and synchronization affect the next physical state. The ongoing 3GPP study on 6G use cases explicitly includes AI-agent communication and network-supported AI/ML training and inference, illustrating this shift from generic connectivity toward AI, data, and computing service exposure [1].

Prior work has largely examined the three stages in separate literatures. Autoregressive language and text-image models established scalable token-based generation [2], [3], while MMLU and BIG-bench broadened evaluation beyond a single task [7], [8]. AgentBench evaluates tool-using trajectories [13], and ReAct and Reflexion study interleaved reasoning, acting, feedback, and retry [16], [17]. Embodied research moves evaluation into manipulation and sensorimotor interaction through Meta-World, RT-2, and LIBERO [22]–[24]. In parallel, communication-oriented studies cover mobile-edge LLM inference [32], semantic and token communications [4], [5], and integrated sensing and edge AI [6]. These streams provide essential components, but use different units of evaluation—content, task trajectory, and physical episode—which motivates a cross-stage synthesis.

This transition exposes a mismatch between conventional communication objectives and application value. Maximizing throughput or minimizing average packet delay does not specify whether a missing visual patch hides a grasp point, whether a stale tool response invalidates the remaining plan, or whether jitter destabilizes a fast control loop. Conversely, transmitting all raw data with uniform reliability is often infeasible: an embodied platform may contain multiple cameras and range sensors, tens of joints, and more than one control timescale. The central design question is consequently: *which information must be delivered, to which model or actuator, by what deadline, and with what task-level loss when resources are limited?*

A. Contributions

The survey makes four contributions.

- It synthesizes generative, agentic, and embodied systems through a network-centric comparative framework that jointly classifies model objectives, token sources, interaction patterns, benchmarks, failure consequences, and network operating bottlenecks.
- It maps AI-side evaluation to network-side quality of service (QoS). The resulting metric ladder progresses

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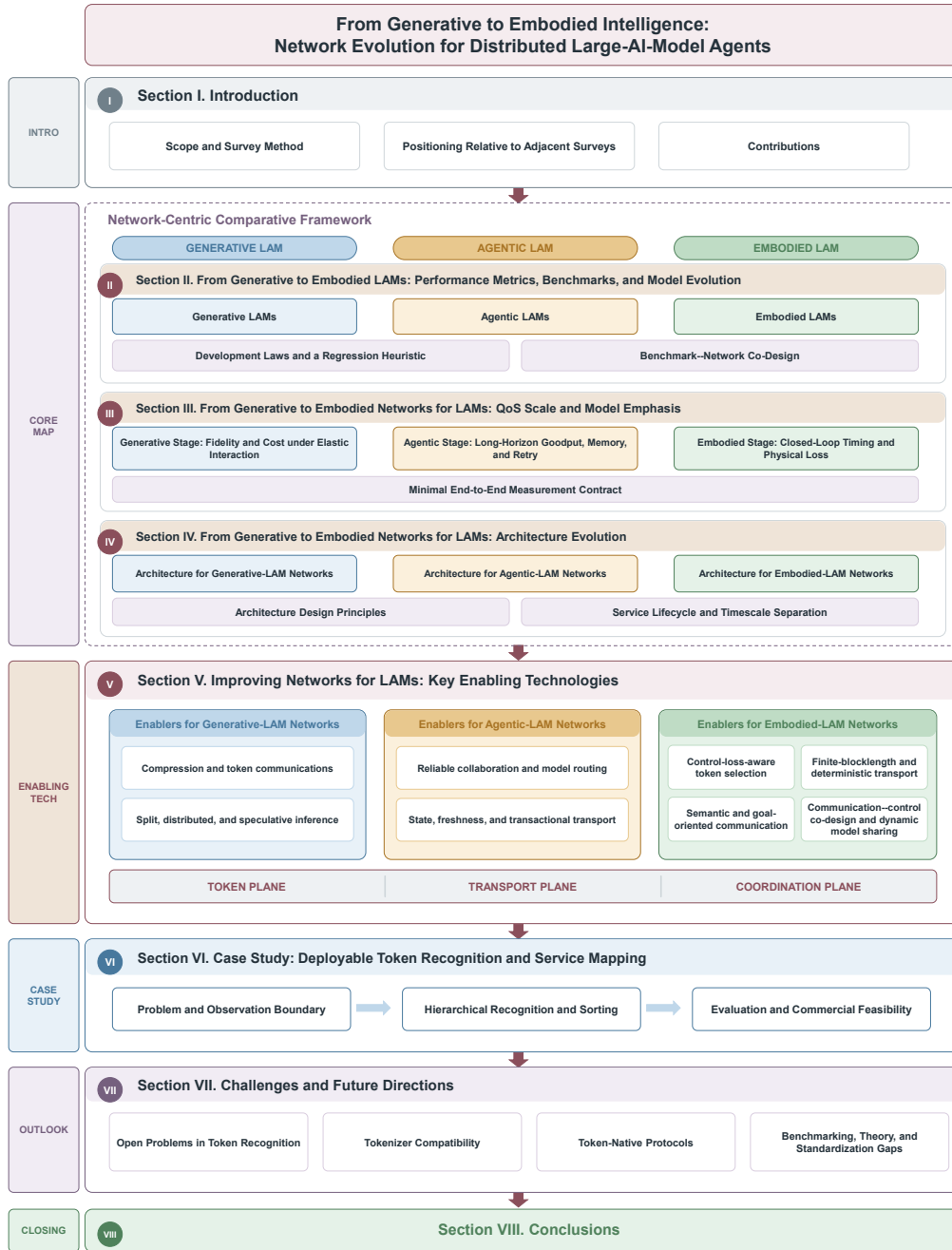


Fig. 1. The outline of this survey.

from time to first token and content quality, to transaction success and long-horizon tail latency, and finally to task distortion, robust loop period, and control cost.

- It organizes enabling technologies into token, transport, and coordination planes and maps them back to the QoS and topology of each stage. For embodied agents specifically, the synthesis exposes two transfer gaps: compression is commonly perception-oriented rather than control-oriented, and deterministic transmission is commonly source-independent rather than task-coupled.
- It develops a deployable token-recognition case study and derives research directions for service classification,

interoperability, and protocol design under confidentiality and commercial-boundary constraints.

Section II reviews model evolution and benchmarks. Sections III and IV study metric and architecture evolution, respectively. Section V surveys enabling technologies, Section VI presents the case study, and Section VII discusses open challenges.

II. FROM GENERATIVE TO EMBODIED LAMs: PERFORMANCE METRICS, BENCHMARKS, AND MODEL EVOLUTION

A. Generative LAMs

Generative LAMs learn a conditional distribution over token sequences [2], [3]. Text is represented by subword tokens; images and video by spatial or spatiotemporal patches or discrete latent codes; and audio by waveform, spectral, or codec tokens. Transformer attention supplies a shared processing interface, but it does not make these tokens computationally equivalent. A text token may represent a word fragment, whereas a visual token represents a local patch and an audio token represents a time interval.

Model-side evaluation in this stage is largely *content-centric*. Perplexity measures predictive fit [2]; exact-match or multiple-choice accuracy measures knowledge and reasoning [7]; and BLEU and ROUGE compare generated and reference text [9], [10]. MMLU spans 57 subjects and tests broad academic knowledge [7]; BIG-bench uses more than 200 heterogeneous tasks to probe capabilities and scaling behavior [8]. For open-ended and multimodal generation, human preference, learned reward, semantic similarity, factuality, and safety are required because literal word or phrase matching cannot capture meaning, visual quality, or factual correctness.

The token workload depends strongly on both modality and direction. Text understanding performs a parallel prefill over the prompt and retrieved context, whereas text generation is autoregressive: it adds one token per decoding step and grows the key-value cache with sequence length. Image understanding processes a spatial patch grid [11], while diffusion-based image generation repeatedly evaluates a denoising model [12]. Video adds a temporal token dimension to image processing, so its cost scales with frame count as well as resolution; audio understanding and generation operate on dense time-indexed spectral or codec tokens and often require causal streaming. Consequently, long text output is dominated by sequential decoding, high-resolution image synthesis by repeated denoising, and video generation by the largest combined compute and memory footprint. These modality-dependent workloads define the relevant local model characteristics in this section.

B. Agentic LAMs

An agentic LAM places a large AI model inside a feedback loop with a planner, tool interfaces, memory, verification, and sometimes other agents. The unit of work is no longer one output sequence but a trajectory

$$\tau = (o_0, a_0, o_1, a_1, \dots, o_K),$$

where observations o_k may be web pages, application screenshots, database results, code-execution logs, or messages from peer agents, and a_k may call an API or change external state. Reasoning techniques such as chain-of-thought and tree/graph search, retrieval-augmented generation, model context protocols, and multi-agent debate increase capability by introducing more rounds and more state.

Consequently, evaluation shifts from answer similarity to *process-grounded task completion*. AgentBench, for example, evaluates LLM agents across operating-system, database, knowledge-graph, gaming, household, web-shopping, and web-browsing environments [13]. WebArena tests goal completion in realistic websites [14], whereas SWE-bench evaluates repository-level issue resolution with executable tests [15]. Across these benchmarks, relevant metrics include task success rate, number of tool calls, constraint violations, plan length, recovery from tool failure, long-horizon consistency, and cost per completed task. An agent that produces fluent intermediate text but executes the wrong transaction has failed.

Trajectory reliability compounds over the sequence. If step k succeeds with conditional probability p_k , a simplified serial workflow has

$$P_{\text{task}} = \prod_{k=1}^K p_k, \quad P_{\text{fail}} \leq \sum_{k=1}^K (1 - p_k), \quad (1)$$

where the equality assumes that each p_k already conditions on prior success. Even modest per-call error becomes material for large K . Reason-act traces and verbal feedback enable exception handling and retry [16], [17], but repeated attempts expand context, consume tokens, and create a heavy tail in job completion time. Agentic-LAM evaluation should therefore report the distribution of attempts, total model and tool calls, recovery success, and terminal task outcome, rather than treating each call as an independent prediction.

C. Embodied LAMs

Embodied LAMs ground language and vision in state estimation, world prediction, planning, and action. Architectures include VLA models, latent world models, bird's-eye-view transformers, and joint-embedding predictive architectures. The observation o_t can combine RGB/depth images, point clouds, force/torque, inertial measurements, joint encoders, and messages from other robots. The output u_t can contain waypoints, discrete skills, joint targets, or continuous control actions. Generalist humanoid models such as GR00T N1 illustrate a trend toward shared representations across embodiments and tasks [21].

The benchmark moves into a physical or high-fidelity simulated loop. Meta-World contains 50 robotic manipulation tasks and evaluates multi-task generalization [22]; RT-2 uses thousands of real-robot trials to test seen and unseen skills and semantic generalization [23]; and LIBERO measures transfer and robustness across lifelong manipulation task suites [24]. VLA evaluations accordingly report task success, trajectory efficiency, collision rate, action error, robustness to scene perturbation, and sim-to-real degradation. World models are judged by predictive consistency and planning quality, not merely reconstruction. Because a small perception error is harmless in one state and catastrophic near a contact transition, average token distortion alone is inadequate.

Safe reinforcement learning formalizes episodic return together with safety constraints or risk penalties [25]. Following

that established evaluation pattern, a benchmark-defined control cost can be written as

$$J_c = \mathbb{E} \left[\sum_{t=0}^H \gamma^t (\|x_t - x_t^*\|_{\mathbf{Q}}^2 + \|u_t\|_{\mathbf{R}}^2 + \lambda_s \mathbf{1}\{\text{unsafe at } t\}) \right], \quad (2)$$

where x_t is the physical state, x_t^* the task-dependent target, u_t the action, and $\mathbf{1}\{\cdot\}$ is an indicator that penalizes safety constraint violation. This is not proposed as a universal new score: each benchmark must publish its horizon, weights, and safety definition and should also report the component metrics separately.

Embodied LAM workloads contain continuous, coupled perception and action streams. The model must understand scene semantics, object relations, and task intent from rich multimodal observations while updating actions in real time; adding more frames or viewpoints improves grounding but increases token count and inference cost. This accuracy–responsiveness tension is difficult, but the streams are not irreducible. Visual tokens contain *spatial* redundancy because only a subset describes task-relevant objects, and action tokens contain *temporal* redundancy during stable motion while becoming highly sensitive near transitions. FlashVLA exploits visual-token pruning and action reuse, reporting substantial compute and latency savings with a small success-rate reduction on LIBERO [29]; algorithm–architecture co-design provides another route to real-time embodied inference [30]. These results motivate exploiting redundancy, while also showing why small corrective actions near an unstable equilibrium cannot be removed indiscriminately.

D. Development Laws and a Regression Heuristic

Table I reveals four recurring development laws. First, evaluation moves from *relative content accuracy*, to *practical process success*, to *physical closed-loop performance*. Second, interaction frequency and coupling increase: a prompt is weakly coupled to the next request, an agent trajectory is logically coupled across steps, and an embodied trajectory is both logically and dynamically coupled through the environment. Third, the consequence of failure changes from a correctable response to an external state change and finally to a potentially irreversible physical event. Fourth, the quantitative workload grows through longer contexts, more model invocations, and denser multimodal streams, whereas the dominant qualitative model concern changes from generation fidelity, to trajectory correctness and memory, and then to grounded control and safety.

The increasing structure of model inputs and outputs creates an opportunity as well as a burden. Generative tokens mainly expose statistical redundancy; agentic trajectories add repeated context and workflow redundancy; embodied streams add constraints induced by geometry, kinematics, and task relations. Selective computation, caching, pruning, and action reuse can exploit these regularities, but their errors must be evaluated at the content, task, or physical-episode level appropriate to the stage.

An intuitive, but deliberately non-normative, interpretation is a *regression heuristic*. Artificial systems have scaled capa-

bilities in an order that is roughly opposite to the layering of human competence: codified knowledge and content generation first, explicit planning and tool use next, and robust sensorimotor interaction last. Conversely, loss of higher explicit reasoning can leave basic procedural or reflexive behavior intact. The analogy helps explain why generating fluent text does not imply reliable tool use, and why either capability does not imply safe physical intelligence. It is neither a biological law nor an originality claim; its engineering implication is only that later stages inherit earlier representation and generation abilities while adding state, timing, interaction, and physical grounding.

E. Benchmark Design Across Stages

A benchmark supports a valid cross-stage comparison only when its execution semantics are preserved. Three mappings are required. First, the benchmark unit must be declared. For generation it may be a prompt and response; for an agent it is a complete trajectory including tool side effects; for a robot it is an episode with initial state, perturbations, and termination criteria. Reporting per-call latency for an agent or per-frame accuracy for a robot silently changes the evaluation unit and can reverse conclusions.

Second, stressors must be injected at the model interface under study. Generative evaluation should vary context length, output length, prompt ambiguity, and modality corruption; agentic evaluation should inject tool failure, stale or missing memory, and misleading intermediate observations; embodied evaluation should vary viewpoint, occlusion, object layout, temporal subsampling, and action noise. Temporal structure must be preserved: random frame removal is not equivalent to losing a contiguous observation interval at grasp contact.

Third, the benchmark must propagate a stressor to the terminal metric. For a generative model this requires scoring the decoded output, not only an internal feature error. For an agent it requires executing the remaining task graph, because one wrong observation can alter later tool calls. For an embodied agent it requires closing the physics loop; isolated perception accuracy cannot show whether the resulting action succeeds safely.

The mapping also reveals a reproducibility hierarchy. A *trace replay* uses fixed model inputs and outputs and is inexpensive, but cannot represent feedback. An *interactive environment* executes tools or a simulator against live model decisions. A *hardware-in-the-loop* setup adds real sensors or actuators and captures timing and embodiment effects. A *field trial* tests distribution shift and safety procedures. Claims should match the highest validated level. In particular, a simulation gain in task success is evidence for an orchestration principle, not proof of real-world safety.

Finally, benchmark scores should be reported as a surface rather than a single operating point. The surface relates model scale, token budget, inference steps, compute load, and task quality. It exposes Pareto fronts and phase transitions—for example, a stable range in which action reuse is safe followed by a sharp loss near contact. Such a surface is the empirical counterpart of the model and task tradeoffs that later system sections must support.

TABLE I
MODEL AND WORKLOAD EVOLUTION ACROSS LAM STAGES

Stage	Primary objective	Token/source structure	Representative evaluation	Interaction/workload pattern	Failure consequence
Generative	Produce useful and faithful digital content	Subwords, image/video patches, audio or codec tokens; statistical and cross-modal redundancy	Perplexity, accuracy, MMLU, BIG-bench, BLEU/ROUGE, preference, factuality	Prompt/context prefill followed by autoregressive decoding; modality-dependent token count; one/few-shot interaction	Hallucinated or degraded content; often detectable, retryable, and non-physical
Agentic	Complete a goal through reasoning, tools, memory, and collaboration	Generated tokens plus UI frames, logs, API objects, retrieved memory, and inter-agent messages; workflow-induced redundancy	Task success, tool correctness, constraint violations, calls/steps, recovery, cost per completed job	Stateful multi-round graph with fan-out, retries, persistent context, and long-tail completion time	Wrong external operation or cascading plan failure; compensation may be expensive or impossible
Embodied	Perceive, predict, and act safely in the physical world	Multiview/multimodal perception, world-state and action tokens; spatial, temporal, task, and physical coupling	Task success, action/trajectory error, collision/safety, world-model error, planning quality, sim-to-real gap, control cost	Continuous, bidirectional, multisource loops at heterogeneous periods; scene understanding and action updates at heterogeneous frequencies	Unstable motion, equipment damage, or injury; late information can be worse than missing information

III. FROM GENERATIVE TO EMBODIED NETWORKS FOR LAMS: QOS SCALE AND MODEL EMPHASIS

The three stages do not require three unrelated sets of network metrics. Latency, bit and token rates, traffic and connection densities, reliability, compute, storage, and energy remain the common engineering quantities. What changes is primarily their *order of magnitude* and the model-side loss that qualifies delivered traffic as useful. Generative models emphasize multimodal fidelity and decode efficiency; agentic models emphasize trajectory success, retries, and long-term memory; embodied models emphasize synchronized state estimation, control, and safety. Table II gives a reproducible scenario envelope that instantiates the radar-style comparison motivating this survey. The entries are design anchors, not standardized 6G requirements or simultaneous maxima.

Two cautions are essential. First, multiplying every maximum in one row by every maximum in another creates an impossible workload; duty factors and statistical multiplexing must be declared. Second, a “token” is not a fixed number of bits across text, images, video, latent states, or actions. Token rate therefore complements rather than replaces bit rate. The assumptions in Table II make sensitivity analysis possible: changing r_0 , N_{par} , L_{state} , a , or trajectory length H immediately produces a new deployment-specific envelope.

A. Generative Stage: Fidelity and Cost under Elastic Interaction

Generative serving decomposes delay into request admission, prefill, and decode. Operational metrics include time to first token (TTFT), time per output token (TPOT), completion time, token/bit goodput, and deadline attainment. Multimodal diversity requires scalable acquisition and fusion, while generation quality limits aggressive token removal: adaptive video tokenization, for example, explicitly trades sequence length against reconstruction and generation quality [31]. Computation is strongly asymmetric between prefill and autoregressive

decode, so batching, cache placement, and the device–edge–cloud split can dominate radio time [32].

Let q_i be response quality, T_i^f TTFT, r_i^{tok} useful output rate, and e_i the communication–compute–storage cost. A generic service utility is

$$U_i^{\text{gen}} = w_q q_i - w_f [T_i^f - d_i^f]^+ + w_r \log r_i^{\text{tok}} - w_e e_i, \quad (3)$$

where $[z]^+ = \max(z, 0)$. Interactive text emphasizes TTFT/TPOT, offline media emphasizes completion energy, and high-fidelity generation emphasizes semantic quality and sustained rate. A token-responsive cost-effectiveness metric is therefore more informative than raw throughput or bits/J alone [33].

B. Agentic Stage: Long-Horizon Goodput, Memory, and Retry

An agentic model autonomously plans, decomposes, reflects, retrieves memory, and invokes tools [34]. Every model response is an intermediate event, so useful network goodput is the fraction of transmitted state that contributes to a correctly completed task. End-to-end job time, task success, recovery probability, context freshness, unnecessary calls, and cost per completed task qualify the same underlying latency, reliability, rate, storage, and energy metrics. Branching, timeouts, debate, and retries produce long tails and repeated token generation/transmission; persistent personalization further turns memory movement and cache affinity into network requirements [35].

A packet may arrive correctly while a stale observation, duplicate non-idempotent tool call, incompatible schema, or poisoned memory causes task failure. Conversely, a lost low-value rationale need not be retransmitted when a verified tool result is intact. A useful transaction-level SLO is

$$S_{\text{ag}} = (P_{\text{task}}, T_{95}^{\text{job}}, A_{\text{mem}}, C_{\text{job}}, P_{\text{unsafe}}),$$

where A_{mem} is the age of relevant memory and C_{job} includes communication, inference, storage, and tool access. Network-aware model-context platforms expose capability and path

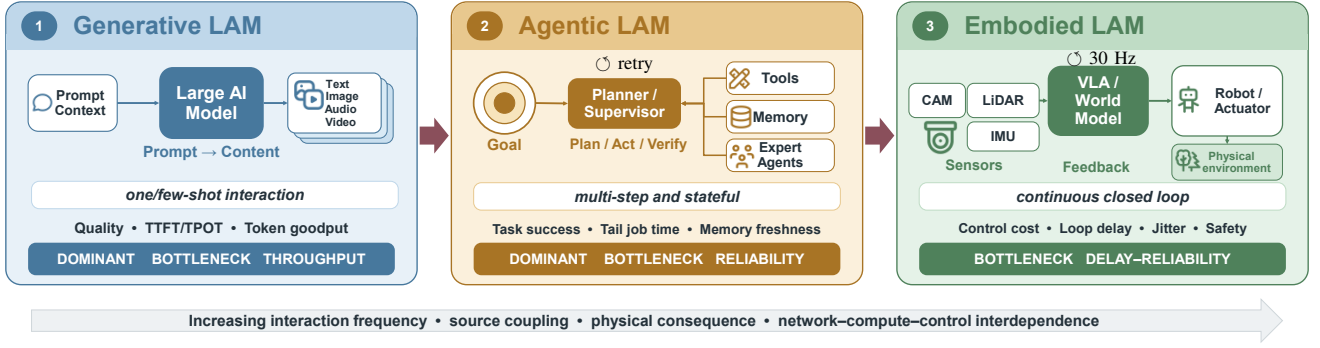


Fig. 2. Evolution of LAM services and their dominant network operating bottlenecks. Requirements accumulate rather than replace one another: embodied systems still need throughput and semantic reliability, but must add deterministic closed-loop behavior. The circular marker on the planner/supervisor denotes retry; each embodied sensing-action loop carries a task-specific update rate, with 30 Hz shown as an illustrative requirement.

TABLE II
ORDER-OF-MAGNITUDE QOS ANCHORS AND EXPLICIT SCENARIO BASIS

Common network quantity	Generative	Agentic	Embodied	Interpretation/calculation basis
Network contribution to latency	100 ms	10 ms	1 ms	Illustrative budget after separating model execution: interactive response, multi-step coordination, and fast physical loop, respectively. Full end-to-end latency still follows (4).
Peak service-token rate	10^3 token/s	5×10^4 token/s	10^5 token/s	$R_{\text{tok}} = N_{\text{par}} r_0$, with source-equivalent stream rate $r_0 = 10^3$ token/s and parallel context/agent/modality counts $N_{\text{par}} = 1, 50, 100$.
Peak service bit rate	0.5 Gbit/s	5 Gbit/s	10 Gbit/s	A high-fidelity multimodal stream is the 0.5-Gbit/s unit; agentic fan-out and embodied multiview sensing aggregate roughly 10 and 20 such units. Compression quality remains model- and modality-dependent [31], [32].
Active-token density	10^7 token/km ²	10^8 token/km ²	10^9 token/km ²	$D_{\text{tok}} = N_c L_{\text{state}}$ using $L_{\text{state}} = 10^3$ active tokens per endpoint and the connection-density anchors below.
Area traffic density	10 Tbit/s/km ²	100 Tbit/s/km ²	1000 Tbit/s/km ²	$D_b = 10^{-3} N_c m a r_b$, where r_b is in Gbit/s. The tuples (m, a, r_b) are (2, 1, 0.5), (1, 0.2, 5), and (1, 0.1, 10); m and a denote parallel flows and active duty factor.
Step/deadline reliability	99%	99.9%	99.999%	$p_{\text{step}} \geq P_{\text{task}}^{1/H}$: $(P_{\text{task}}, H) = (0.99, 1)$, $(0.99, 10)$, and $(0.999, 100)$ yield approximately the listed anchors.
Connection density	10^4 /km ²	10^5 /km ²	10^6 /km ²	Tenfold scenario scaling from personal generative sessions, through dense agent/tool endpoints, to massive sensors and physical agents; the top anchor matches the established massive-connectivity order [1].

state [36]; representation and cache exchange can reduce repeated text and prefill traffic [37], [38], but admission, idempotence, rollback, and audit still apply to the complete task graph.

C. Embodied Stage: Closed-Loop Timing and Physical Loss

Embodied intelligence inherits generative fidelity and agentic trajectory reliability, then adds sensing, actuation, and a physical deadline:

$$T_t^{\text{loop}} = T_t^{\text{sense}} + T_t^{\text{enc}} + T_t^{\text{ul}} + T_t^{\text{queue}} + T_t^{\text{inf}} + T_t^{\text{dl}} + T_t^{\text{act}}. \quad (4)$$

Reducing radio time while increasing encoder or accelerator delay can worsen the loop. For a loop period h_i , jitter tolerance ϵ_i , and violation probability δ_i , a robust SLO is

$$\Pr\left\{T_{i,t}^{\text{loop}} \leq h_i, \left|T_{i,t}^{\text{loop}} - \mathbb{E}T_{i,t}^{\text{loop}}\right| \leq \epsilon_i h_i\right\} \geq 1 - \delta_i. \quad (5)$$

The first condition controls deadline reliability and the second jitter. Asynchronous modalities also require age and skew bounds: a perfectly delivered camera frame from before a contact transition can be less useful than a lower-resolution but current frame.

For retained token set $\mathcal{K}$, goal g , and state x , define the task-conditional distortion

$$D_{\text{task}}(\mathcal{K} | g, x) = J_c(\mathcal{K} | g, x) - J_c(\mathcal{K}_{\text{full}} | g, x). \quad (6)$$

The same token can be dispensable in free space and critical near an obstacle. The corresponding economic metric should measure control improvement per total resource cost:

$$\eta_{\text{EAI}} = \frac{J_c^{\text{baseline}} - J_c^{\text{scheme}}}{\alpha B + \beta E + \chi F + \zeta S}, \quad (7)$$

where B, E, F, S denote bandwidth, energy, compute cycles, and storage. This form discourages spending large resources on tokens that do not improve the physical task.

D. Minimal End-to-End Measurement Contract

The operator need not observe prompts, plans, sensor content, or actions. A commercially plausible contract keeps rich semantics inside the trusted endpoint/vertical and exposes only an authenticated service identifier, generation time, coarse deadline/reliability bin, chunk size, and an opaque causal identifier. Endpoint and compute-domain telemetry can privately bind the identifier to model queues and outcomes. Thus the carrier can attribute radio, path, and edge-queue delay without deep packet inspection or direct access to upper-layer state.

Measurement granularity scales with risk: per-request aggregation for best-effort generation, per-step spans for an agentic transaction, and per-loop timing for an embodied service. Results should report percentiles and violation runs, condition on load, separate offered/decoded/useful goodput, and report joint freshness for fused modalities. For modalities $m \in \mathcal{M}_i$, a useful online proxy is

$$G_i^{\text{sync}} = \frac{1}{|\mathcal{T}|} \sum_{k \in \mathcal{K}_i} v_k \mathbf{1} \left\{ t_k^{\text{rx}} \leq d_k, \max_{m,n \in \mathcal{M}_i} |a_m - a_n| \leq s_i \right\}, \quad (8)$$

where v_k is an endpoint-assigned value bin, a_m information age, and s_i permitted skew. This metric is auditable network telemetry, not a replacement for application-owned quality or safety evaluation.

The records close an assurance loop: admission predicts resource distributions, runtime monitoring detects joint violations, endpoints adapt compression, placement, or period, and aggregated outcomes update the next admission model. This limited exposure is sufficient for service-level engineering and is more deployable than assuming that a network can interpret raw tokens.

IV. FROM GENERATIVE TO EMBODIED NETWORKS FOR LAMS: ARCHITECTURE EVOLUTION

Architecture evolves along three coupled axes: *placement* across device, edge, and cloud; *network topology* among models, users, tools, and sensors; and *collaboration mode* among agents. Centralization is not a binary choice. One-to-one offloading, many-to-one aggregation, one-to-many dispatch, peer-to-peer exchange, and hierarchical hybrids coexist and should be selected by task dependency, latency, trust, and physical coupling.

A. Architecture for Generative-LAM Networks

A generative-LAM network is commonly cloud-centric. User equipment performs tokenization and lightweight preprocessing; an edge gateway provides admission, caching, privacy filtering, or small-model inference; and a cloud cluster hosts the full model. Model, tensor, pipeline, and expert parallelism distribute inference inside or across clusters. Wireless distributed mixture-of-experts systems additionally make expert routing dependent on channel and compute heterogeneity [39]. Pipeline parallelism and batching trade communication bubbles against accelerator utilization [40].

The dominant topology is one-to-one user-model access over a many-to-one serving hub. Users collaborate indirectly by sharing batches, model replicas, and KV-cache infrastructure, rather than by exchanging task state. The edge absorbs latency-sensitive or private work and the cloud supplies model scale; this cloud-edge-device continuum is the basic mobile-edge-intelligence architecture for LLMs [32].

The main control loop is a slow resource-management loop: predict request and token lengths, place models, reserve accelerators and backhaul, batch compatible requests, and route overflow. Split inference places early layers on the device and later layers at the edge, transmitting intermediate features.

Progressive feature transmission allows a receiver to stop once confidence is sufficient [41]. This architecture is effective when the user transaction is largely independent and deadlines are elastic. It becomes inefficient when every intermediate action depends on fresh tool or environment state.

B. Architecture for Agentic-LAM Networks

Agentic architecture adds a stateful orchestration plane. A planner or supervisor discovers capabilities, selects models and tools, maintains a task graph, and commits results to short- and long-term memory. Executors can be distributed over devices, edge sites, enterprise services, and clouds. Messages carry not only tokens but role, goal, schema, provenance, confidence, and dependency. Multi-agent systems use debate, vote, expert routing, or hierarchical decomposition; the best topology depends on task structure and the correlation of model errors [18], [19], [42]. One-to-many dispatch appears when a planner fans out to tools or specialists; many-to-one aggregation appears in voting and fusion; peer-to-peer exchange supports decentralized debate; and a supervisor plus peer subgraphs forms a hybrid. Emerging MCP, ACP, A2A, and ANP families cover different discovery, tool-access, messaging, and decentralization assumptions [43]. Thought- or latent-state exchange may further change the payload while retaining the same task graph [44], [45].

This design resembles a distributed transaction system but has probabilistic components. It requires:

- 1) capability and context discovery before routing;
- 2) session affinity for key-value caches and private memory;
- 3) checkpoint, retry, and idempotence semantics for tool calls;
- 4) verification paths for high-impact actions; and
- 5) joint compute-network-storage admission for the complete task rather than each RPC.

The architecture may keep a small model on-device for privacy and rapid screening, escalate uncertain inputs to a remote large model, and invoke specialists only when their expected information gain exceeds their latency and cost. Uncertainty-aware hybrid inference is one realization of this principle [46].

C. Architecture for Embodied-LAM Networks

Embodied systems require a hierarchy of loops (Fig. 3). A local reflex/control plane executes high-rate stabilization and safety interlocks. An edge cognition plane performs multi-modal fusion, VLA inference, local mapping, and coordination with deterministic access. A cloud learning plane provides large-model reasoning, fleet memory, simulation, and non-real-time model updates. A slow cloud model should not directly occupy a hard real-time loop; it supplies goals, policies, latent plans, or verified skill updates to an edge/local controller. Decentralized local safety and reflexes coexist with temporarily centralized edge fusion. Consequently, collaboration becomes dynamic and partially centralized rather than globally centralized or fully peer-to-peer.

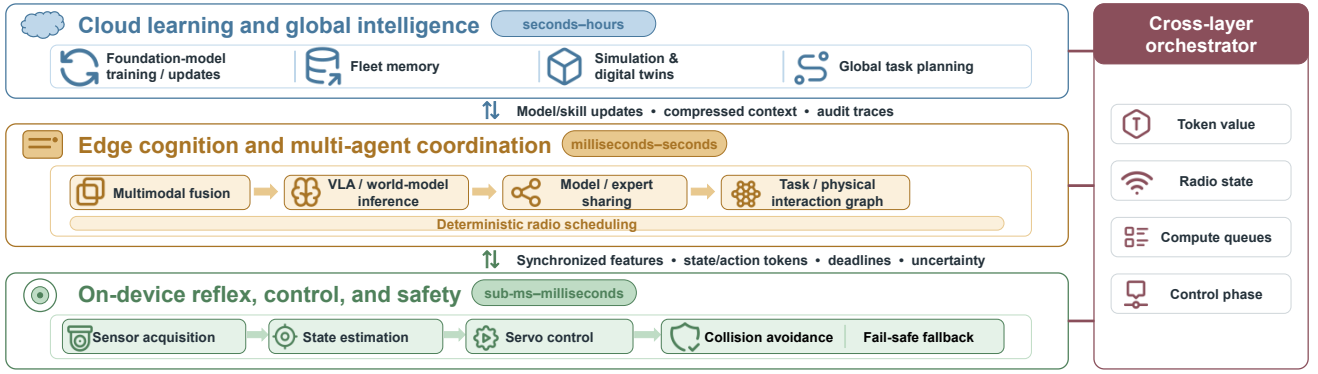


Fig. 3. Cloud-edge-device hierarchy for distributed LAM services. The generative stage primarily uses cloud scale, the agentic stage adds stateful edge orchestration, and the embodied stage keeps hard safety and stabilization local while the edge closes latency-critical cognition and coordination loops.

The architecture is also *partially centralized*. If two robots have neither task dependence nor physical interference, merging their observations in one model can add quadratic attention cost without useful information. When they jointly carry an object or occupy the same collision region, shared inference can improve consistency and reduce duplicated sensing. Let $\mathcal{G}_t = (\mathcal{V}, \mathcal{E}_t)$ be a time-varying interaction graph with

$$w_{ij}(t) = \rho_{\text{task}} I_{ij}^{\text{task}}(t) + \rho_{\text{phy}} I_{ij}^{\text{phy}}(t) + \rho_{\text{info}} I_{ij}^{\text{info}}(t), \quad (9)$$

where the three terms quantify task dependency, physical interference, and complementary information. Clustering or sparsifying $\mathcal{G}_t$ determines which agents share a model or exchange tokens. The resulting groups can run at different loop periods, producing an asynchronous, multi-rate networked control system.

D. Architecture Design Principles

Three principles follow. First, *place deadlines before models*: determine which loop can tolerate cloud, edge, or only local execution before selecting model size. Second, *centralize only useful mutual information*: shared models should be activated by task or physical coupling rather than by agent density alone. Third, *retain a safe degraded mode*: if the edge model, radio link, or clock synchronization fails, the device must fall back to a local controller that maintains a safe set, even if task progress stops.

E. Service Lifecycle and Timescale Separation

Architecture must cover the service lifecycle, not only the inference data path. During *registration*, an endpoint or vertical gateway publishes an authenticated coarse capability/SLO manifest; detailed model state, sensor content, action semantics, and safety policy remain inside its trust domain. During *admission*, the orchestrator maps a requested service to a service graph, predicts complete resource demand, selects placements, and rejects or degrades a task whose robust SLO is infeasible. During *execution*, fast schedulers allocate token, radio, and compute resources under the admitted envelope. During *recovery*, endpoints switch models, paths, or controllers and preserve transaction semantics. During *learning*,

traces update workload, token-value, channel, and physical interaction models.

The lifecycle contains at least three control timescales.

- At a *planning timescale* (minutes to hours), models and adapters are placed, capacity and slices are reserved, and fleet-wide policies are updated. Forecast error dominates.
- At a *task timescale* (hundreds of milliseconds to minutes), agents/tools are selected, model-sharing groups and loop periods change, and context migrates. Task and physical topology dominate.
- At a *loop/slot timescale* (sub-millisecond to tens of milliseconds), features and actions are scheduled, short packets are coded, and compute slots are aligned. Channel, queue, and release-time uncertainty dominate.

Optimizing every variable at the fastest timescale is neither feasible nor stable. Slow decisions should pass resource and risk budgets downward; fast controllers should return feasibility margins and violation statistics. When fast scheduling cannot maintain an admitted envelope, it should trigger a defined degradation—smaller model, fewer views, slower noncritical loop, or local safe mode—rather than silently accumulating queue.

Mobility makes state migration a first-class operation. A generative session can move after transferring a key-value cache, subject to a short pause. An agentic session also needs memory and tool-transaction state. An embodied session cannot pause an unsafe plant while a large cache migrates. Predictive handover should replicate only the state needed to continue the near-term loop, keep stabilization local, and transfer large context in the background. Multi-tier computation offloading [47] and ultra-low-latency edge inference [48], together with reliability-aware edge offloading [49], provide resource-allocation foundations, but embodied migration adds continuity of action and safety state.

Finally, administrative and physical trust domains need not coincide. A warehouse owner, robot vendor, network operator, and cloud-model provider may control different lifecycle stages. The architecture therefore needs signed capability manifests, explicit data-use policy, model provenance, and auditable responsibility at every boundary. The vertical remains responsible for translating private semantics and safety

TABLE III
PLACEMENT, NETWORKING TOPOLOGY, AND COLLABORATION EVOLUTION

Stage	Cloud-edge-device placement	Typical topology	Collaboration law	Persistent state/coupling	Timescale
Generative	Device front end; edge gateway/cache or small model; cloud accelerator cluster	One-to-one request path over a many-to-one serving/cache hub; occasional split pipeline	Resource sharing dominates direct agent cooperation: admission, batching, cache-aware routing, and expert placement	Request, model/tokenizer version, KV cache, queue and radio-compute utilization	Request and decode
Agentic	Device supervisor; edge session/state broker; distributed cloud/edge models, tools, and stores	One-to-many planner dispatch, many-to-one vote/fusion, peer debate, or supervisor-peer hybrid	Collaboration follows the task graph; central orchestration gives way to decentralized/hybrid execution when trust, failure isolation, or scale requires it	Goal, dependencies, memory/context version, tool side effects, provenance; network-compute-storage transaction	Step and workflow
Embodied	Local reflex/safety; edge VLA/world-model fusion; cloud fleet learning and simulation	Dynamic many-to-many interaction graph with local peer links and temporary edge-centered clusters	Partial centralization follows task and physical coupling; local safety persists while shared inference is activated only for interacting agents	Multimodal time/age, action history, uncertainty, safety envelope; sensing-communication-computing-control coupling	Sub-ms-hours

policy into coarse service classes; the operator authenticates, meters, and polices those classes without requiring raw-token visibility.

V. IMPROVING NETWORKS FOR LAMS: KEY ENABLING TECHNOLOGIES

We group mechanisms into three interacting planes. The *token plane* decides what representation to compute and communicate. The *transport plane* decides how to deliver it by a deadline. The *coordination plane* decides which agents/models should share information and how network and compute resources close the loop. These planes are not a separate taxonomy: Table IV maps each one to the common QoS scale in Table II and the placement/topology in Table III.

A. Enablers for Generative-LAM Networks

1) *Compression and token communications*: Quantization, structured pruning, low-rank adaptation, key-value-cache compression, and top- k /top- p decoding reduce memory, computation, or output entropy. Token communications elevates tokens to semantic communication units and uses cross-modal context to reconstruct masked or corrupted content [5]. Compared with conventional source coding, a receiver equipped with a compatible foundation model can exploit learned priors. This can reduce payload but transfers burden to shared model state, tokenizer agreement, uncertainty calibration, and semantic error detection. Token-domain multiple access further lets many devices share tokenizer and modulation codebooks, using contextual reconstruction to mitigate token collisions [50]. It targets the high token and connection densities in Table II, but its gains depend on codebook compatibility and on whether reconstructed content satisfies the generative quality target.

For sources $Z_1, \dots, Z_M$ and task goal g , the relevant multi-source rate-distortion function is conceptually

$$R_g(D) = \min_{p(\hat{z}|z_1, \dots, z_M)} I(Z_1, \dots, Z_M; \hat{Z} | g), \quad (10)$$

$$\text{s.t. } \mathbb{E}[D_{\text{task}}(Z, \hat{Z} | g)] \leq D.$$

Unlike pixel mean-square error, D_{task} can be nonconvex and state-dependent. Cross-modal and multiview redundancy are not additive: removing a camera token may be harmless when lidar observes the same object but damaging when glare invalidates the lidar-camera correspondence.

2) *Split, distributed, and speculative inference*: Split inference selects a cut layer and transmits intermediate features. Tensor parallelism distributes layer operations and repeatedly aggregates partial results; over-the-air computation can exploit wireless superposition to accelerate aggregation [51], [52]. Sparse-feature over-the-air fusion similarly combines multi-view features before reconstruction or inference [53]. These schemes are attractive when synchronized devices observe a common scene, but analog aggregation noise and stragglers must be evaluated at the task output. Intent-aware VLM split computing for disaster response is an early example of extending the placement decision toward embodied context [54].

Speculative decoding lets a small “draft” model propose tokens and a large model verify them, preserving the target model distribution while reducing serial decode cost [55]. Wireless implementations must jointly select the draft model, speculation length, offloading location, and radio resources. A low draft acceptance rate wastes both inference and communication. For generative services, the decision can be driven by expected decode speedup; for embodied services it must also account for action deviation and deadline risk [56].

B. Enablers for Agentic-LAM Networks

1) *Reliable collaboration and model routing*: Multi-model collaboration can improve coverage and verification through expert routing, debate, vote, or sequential refinement. The network problem is to choose the smallest collaboration graph that reaches a target task reliability before a deadline. Correlated models should not be treated as independent replicas: if they share training data or failure modes, majority vote offers less diversity than its communication cost suggests [18], [19].

The message representation is itself an architectural decision. Q-KVComm adaptively quantizes transmitted KV state,

KVComm selectively shares informative layers, and latent-space communication avoids decoding and re-encoding a complete natural-language message [37], [38], [45]. SemShareKV reuses cache entries across semantically similar rather than lexically identical prompts [57]. These results substantiate a shift from repeated surface text toward state reuse, but they do not remove version, privacy, or semantic compatibility constraints.

Adaptive repeat-query systems learn whether another response is likely to move the latent solution toward a correct concept [20]. More generally, an agent should stop when the expected value of another message is lower than its marginal compute, token, and latency cost. Network schedulers can use confidence and disagreement as hints, but final verification must remain application-controlled because confidence is not a safety certificate.

2) *State, freshness, and transactional transport*: Agent traffic requires protocol support for a task identifier, parent step, model/tool version, context hash, timeout, and idempotence policy. Memory and retrieval should be scheduled by *age of relevant information*, not storage age alone. A newer observation can be irrelevant while an older constraint remains binding. For irreversible tools, an “exactly once” semantic cannot be inferred from transport acknowledgments; the tool and orchestrator need a commit protocol and audit log.

Compute-aware admission is equally important. Reserving radio resources for a tool call without an available model replica merely shifts queuing delay. Conversely, prewarming a model without a feasible wireless path wastes accelerator memory. Joint admission can operate at two timescales: slow model placement and slice reservation, followed by fast task-graph scheduling and rerouting. The resulting design corresponds directly to the agentic 10-ms and 99.9% scenario anchors: representation sharing reduces rate/compute load, while debate, heterogeneous retry, and transactional transport raise task reliability at an explicit tail-latency cost.

C. Enablers for Embodied-LAM Networks

1) *Control-loss-aware token selection*: Embodied compression should estimate the marginal increase in control cost caused by removing, quantizing, delaying, or reusing token z_k :

$$s_k(t) = \frac{\mathbb{E}[\Delta J_c \mid z_k, x_t, g_t]}{\Delta b_k + \alpha \Delta f_k + \beta \Delta e_k}, \quad (11)$$

where the denominator is saved communication, compute, and energy resource. Tokens with high s_k receive stronger coding or earlier service. The score changes with control phase: background visual patches can be pruned during free motion, contact geometry becomes critical at grasp time, and action reuse is safest inside stable segments. Channel-wise VLA quantization studies also indicate that channels have unequal action sensitivity [58].

A practical estimator can combine attention/gradient saliency, model uncertainty, object and contact semantics, and counterfactual rollouts in a world model. None alone is sufficient. Attention can be poorly calibrated to causal task value; gradients are local; and rollouts inherit model bias. Bregman

divergences provide a family of representation losses, while task-conditioned rate-distortion analysis connects them to resource allocation. The open theoretical problem is to bound physical control loss when the VLA mapping is nonlinear and the source modalities are dependent.

2) *Finite-blocklength and deterministic transport*: URLLC techniques provide a foundation for short embodied messages. For blocklength n , signal-to-noise ratio γ , and error target ε , the finite-blocklength rate is approximated by

$$R(n, \varepsilon) \approx C(\gamma) - \sqrt{\frac{V(\gamma)}{n}} Q^{-1}(\varepsilon) + \frac{\log_2 n}{2n}, \quad (12)$$

where C and V are channel capacity and dispersion [59]. Finite-blocklength cell-free transmission and joint up-link/downlink allocation further expose hard-deadline and energy tradeoffs for multiple devices [60], [61]. This exposes the bandwidth-latency-reliability tradeoff hidden by asymptotic capacity.

Interface diversity, multi-connectivity, packet duplication, and optimized HARQ reduce outage or retransmission delay [62], [63]. Preemptive puncturing allows urgent traffic to overwrite scheduled broadband resources, while slicing isolates service classes [64]–[66]. Two-timescale QoS-aware multiplexing is useful when slow reservation must anticipate fast allocation under nonstationary demand [67]. Deterministic-delay work further optimizes jitter, hard-deadline violation, and high-order delay statistics under finite-blocklength transmission [68], [69].

However, direct transplantation to embodied traffic has two limitations. First, traditional models often assume independent sources and a fixed utility loss for punctured broadband packets. Multimodal embodied sources are coupled, so simultaneous degradation can remove all views of a critical object. Second, radio delay is only one term in (4). An end-to-end deterministic service must coordinate sensing release times, encoder and accelerator queues, radio slots, receiver inference, and actuation.

3) *Semantic and goal-oriented communication*: Goal-oriented communication schedules information by its effect on a task. Value and age of information have been applied to robotic waypoint delivery [70]. Integrated sensing and edge AI similarly co-designs sensing, communication, and inference around task performance [6]. For embodied LAMs, semantic communication should protect control objectives, cross-modal complementarity, and multiview complementarity. It should also report uncertainty: hallucinated recovery of a missing object is unacceptable when a safety controller interprets it as observation. Rate-limited closed-loop semantic communication offers a direct bridge from task representations to distributed sensing and control [71].

A useful design is *loss-controlled non-orthogonal reuse*. Two sources may overlap in radio resources if their joint task distortion remains below a bound; sources covering the same hazard should instead receive diversity. This reverses a common heuristic: highly correlated data can be aggressively compressed for efficiency, but correlation alone does not justify common-mode failure.

4) *Communication-control co-design and dynamic model sharing*: Wireless networked control studies optimize bandwidth, power, uplink/downlink ratio, sampling period, and control policy jointly. Recent work considers sensing-communication-computing-control loops for robotic digital twins [26], communication-control codesign for large-scale systems [27], and communication allocation for system identification [28]. Energy-aware wireless communication-control co-design further shows that loop performance and radio efficiency should be optimized together [72]. These results show that optimizing physical control or identification can produce allocations very different from throughput maximization.

Most tractable theory assumes linear dynamics and a common period. Embodied LAM systems add black-box nonlinear policies, partial observability, asynchronous periods, and time-varying inter-agent coupling. A robust orchestrator must jointly choose model-sharing groups $\mathcal{C}_t$, periods h_i , token sets $\mathcal{K}_{i,t}$, bandwidth $b_{i,t}$, and compute $f_{i,t}$:

$$\min \sum_i \omega_i J_{c,i} + \lambda_B \sum_i b_i + \lambda_F \sum_i f_i + \lambda_E E, \quad (13a)$$

$$\text{s.t. } \Pr\{T_{i,t}^{\text{loop}} \leq h_i, |\Delta T_{i,t}| \leq \epsilon_i h_i\} \geq 1 - \delta_i, \quad (13b)$$

$$D_{\text{task},i}(\mathcal{K}_{i,t}, \mathcal{C}_t) \leq D_i^{\max}, \quad (13c)$$

$$\sum_i b_{i,t} \leq B, \quad \sum_i f_{i,t} \leq F, \quad (13d)$$

$$\mathcal{C}_t \in \mathcal{C}_{\text{safe}}. \quad (13e)$$

Slow graph clustering and model placement can be separated from fast token, radio, and compute scheduling, but the two timescales must exchange feasibility margins. Distributionally robust or risk-sensitive optimization is promising because empirical delay and control-loss distributions shift with scenes and tasks.

VI. CASE STUDY: DEPLOYABLE TOKEN RECOGNITION AND SERVICE MAPPING

A. Problem and Observation Boundary

Token-aware scheduling first requires the network to identify what service a chunk belongs to. Payload inspection is an unsuitable premise: traffic is normally encrypted, proprietary model state is commercially sensitive, and a carrier cannot be expected to infer application semantics or safety policy. This case study therefore considers a deployable boundary in which a trusted endpoint or vertical-domain gateway converts private state into a small, authenticated descriptor. The carrier observes only conventional flow features and coarse bins; rich meaning remains end to end.

Let $c \in \{G, A, E, U\}$ denote generative, agentic, embodied, or unknown service. The endpoint may provide metadata m consisting of an opaque service ID, chunk/transaction identifier, timestamp, deadline bin, reliability tier, and token-count or byte-count bin. The network can independently observe features x such as directionality, burst/periodicity, inter-arrival variation, chunk size, fan-out/fan-in degree, retry pattern, and edge-queue association. It does not observe the prompt,

chain of thought, tool arguments, sensor content, or action. Recognition is a cost-sensitive decision

$$\hat{c} = \arg \min_{j \in \{G, A, E, U\}} \sum_i C_{ij} \Pr(c = i | x, m), \quad (14)$$

where C_{ij} prices a false promotion as wasted reserved capacity and a false demotion as SLO risk. For an authenticated embodied class, $C_{E,G}$ and $C_{E,A}$ should exceed $C_{G,E}$, but only the responsible vertical—not the carrier classifier—is allowed to assert physical safety.

B. Hierarchical Recognition and Sorting

Recognition operates at three granularities. First, *registration-time recognition* verifies the service manifest, tenant authorization, endpoint attestation, and allowed maximum rate/reliability class. This is slow and commercially enforceable: the operator sells reserved network/edge resources, not knowledge of private tokens. Second, *session-level recognition* combines the signed class with traffic behavior. A mismatch—for example, an allegedly sparse generative session producing a dense periodic load—triggers rate policing or re-admission rather than semantic inspection. Third, *chunk-level sorting* uses deadline, dependency, and value bins already assigned by the endpoint. The network orders chunks inside the admitted envelope but does not compute their task value.

The hierarchy controls overhead. If a compact descriptor adds $h = 8$ bytes to a 1-kbyte chunk, its byte overhead is $h/(1024 + h) = 0.78\%$; attaching it to every small token would be far more expensive. Flow-level metadata is therefore adequate for elastic generation, transaction/chunk metadata is useful for agentic steps, and fine chunking is reserved for deadline-critical embodied transitions. Token-level scheduling is justified only when its marginal SLO gain exceeds header, authentication, and queue-management cost.

Table V maps each recognized class to the quantitative anchors in Table II; implementation replaces those anchors with an admitted tenant profile. For G , the scheduler batches and routes by model/cache while preserving a quality-qualified completion deadline. For A , parent-step and expiry metadata prevent a late rationale from blocking a verified commit, and session affinity avoids repeated memory/KV movement. For E , timestamps and skew groups align sensing, radio, edge compute, and action delivery; the local controller remains the fallback if the admitted envelope cannot be sustained.

C. Evaluation and Commercial Feasibility

Plain classification accuracy is insufficient. A test should report classwise precision/recall, calibration error, descriptor and cryptographic overhead, false-promotion resource waste, false-demotion SLO violations, useful goodput, radio/compute utilization, and cost per completed request/task/loop. Workloads must mix all four classes and include class imbalance, encrypted legacy flows, burst loss, edge saturation, mobility, version mismatch, replayed descriptors, and a tenant that inflates priority. Comparisons should include conventional

TABLE IV
STAGE-BY-STAGE MAPPING FROM QoS AND ARCHITECTURE TO ENABLING TECHNOLOGIES

Stage	QoS/model target	Placement/topology	Token/data-plane enablers	Transport/compute enablers	Coordination enablers and open gap
Generative	100-ms/ 10^3 -token/s anchor; content fidelity, TTFT/TPOT, and cost	Cloud-dominant one-to-one access over many-to-one edge/cloud serving	Adaptive tokenization; token pruning/quantization; TokCom and ToDMA [5], [50]	Batch/cache-aware scheduling; split/progressive inference; speculative decode [41], [55]	Joint radio-compute admission and model/expert placement; open gap is generalization across modalities, tokenizers, and model versions
Agentic	10-ms/5 × 10^4 -token/s/99.9% anchors; task goodput, memory freshness, job tail, and cost	One-to-many tools, many-to-one aggregation, peer debate, or hybrid task graph	KV quantization/selection, semantic cache reuse, machine/latent-state exchange [37], [38], [45], [57]	Transactional partial reliability; cache affinity; early exit, heterogeneous retry, and joint network-compute-storage admission	Capability discovery, minimum reliable debate graph, model routing, and rollback [20], [36]; open gap is reliable interoperability without exposing private plans or memory
Embodied	1-ms/ 10^5 -token/s/99.999% anchors; loop tail, jitter, synchronized goodput, control cost, and safety	Local reflex plus temporary edge-centered clusters on a dynamic many-to-many interaction graph	Control-loss-aware token/channel selection, task-conditioned compression, semantic/goal-oriented updates	Finite-blocklength coding, diversity/HARQ, slicing, deterministic radio-compute scheduling, and safe local fallback	Interaction-graph clustering, multi-rate model sharing, and communication-control co-design [26], [27]; open gap is a bound from nonlinear token error to long-horizon physical loss

TABLE V
TOKEN/CHUNK RECOGNITION EVIDENCE AND CONSERVATIVE NETWORK ACTIONS

Class	Trusted endpoint evidence	Observable traffic tendency	QoS/model qualification	Network mapping and fallback
Generative (G)	Request/response ID, model/tokenizer version hash, output deadline and token bin	Bursty context uplink followed by asymmetric autoregressive or media downlink; one/few transactions	100-ms and 10^3 -token/s anchors; qualify rate by content quality, TTFT/TPOT, and cost	Cache/model-affine routing, batching and elastic rate; on missing/invalid metadata use ordinary interactive/media QoS
Agentic (A)	Session plus parent-step ID, expiry, idempotence/commit flag, memory-version hash, and coarse retry tier	Multi-round fan-out/fan-in among model, tool, and memory endpoints; correlated bursts, retries, and long-lived state	10-ms, 5×10^4 -token/s, and 99.9% anchors; qualify by complete-task success, job tail, freshness, and cost	Session affinity and joint radio-compute-storage admission; protect commits, expire stale rationales, and use bounded heterogeneous retry
Embodied (E)	Loop ID, measurement/action timestamp, period/deadline and skew bins, signed local-fallback capability	Periodic or event-triggered multisensor uplink plus small action downlink; correlated sources and strict release times	1-ms, 10^5 -token/s, and 99.999% anchors; qualify by loop-tail, jitter, synchronized goodput, and application-owned control outcome	Align radio and edge-compute slots, reserve/duplicate critical chunks, and keep stabilization local; unverified E declarations are not granted a safety tier
Unknown (U)	No descriptor, failed authentication, or unsupported version	Any encrypted legacy flow; behavior may resemble more than one stage	No semantic or safety claim can be inferred	Conventional QoS and congestion control; never infer a high-impact class solely from traffic shape

five-tuple QoS, traffic-only recognition, authenticated metadata only, and the hybrid in (14).

The business boundary is equally testable. The operator guarantees measurable resources and network-side delay/reliability percentiles; the application or vertical owns model quality, tool correctness, and physical safety. Charging can be based on reserved rate, compute, storage, and verified service-class usage rather than on prompt meaning or private memory contents. If a vendor declines metadata exposure, its traffic remains class U and retains ordinary connectivity. This opt-in, backward-compatible contract is less ambitious than network-side semantic interpretation, but it is implementable across trust and administrative domains.

VII. CHALLENGES AND FUTURE DIRECTIONS

A. Open Problems in Token Recognition

Section VI deliberately stops at endpoint-declared, authenticated classes. The harder research problem is to estimate task value within a trusted model/vertical without turning the carrier into an application controller. Attention, activation magnitude, entropy, or gradient saliency can correlate with value but can fail under distribution shift. Counterfactual estimators ask how task or control cost changes if a token

is delayed, quantized, predicted, or omitted; world-model rollouts can approximate the answer, while conformal or robust bounds express uncertainty. Rare safety-critical events remain the difficult case.

Deployment needs verifiable delegation: the application computes rich value, a trusted gateway maps it to a few tariff/SLO bins, and the operator meters the bins against an admitted envelope. Research should quantify how much gain is lost at each coarser granularity and whether the gain covers authentication, header, scheduling, and dispute-resolution cost. Traffic-only inference may detect abuse, but must never silently promote an encrypted flow to a safety-critical class.

B. Tokenizer Compatibility

Token communications assumes that sender and receiver agree on token boundaries, codebooks, embeddings, and model semantics. In practice, tokenizers differ by model, version, language, modality, and embodiment. Even identical token IDs are meaningless without a versioned vocabulary and embedding interpretation. Visual and action tokenizers are especially unstable because codebooks can change during fine-tuning.

Compatibility has four layers:

- 1) *syntactic*: vocabulary and token-ID mapping;
- 2) *representational*: embedding dimension, quantization, and feature normalization;
- 3) *semantic*: equivalent meaning or task effect across models; and
- 4) *behavioral*: equivalent downstream action within a safety tolerance.

Only the first two can be solved by a schema registry. Semantic and behavioral compatibility require calibration data, adapters, or a shared latent interface. A universal tokenizer may simplify networking but can sacrifice model efficiency and lock the ecosystem to one representation.

Future protocols should negotiate a tokenizer/model manifest containing a cryptographic version hash, modality, vocabulary or codebook identifier, embedding format, supported adapters, calibration range, and fallback codec. Gateways can translate tokens through learned adapters, but translation adds latency and an error that must be included in (6). For safety-critical actions, a receiver should reject an unknown or out-of-range codebook rather than guess.

Model updates create a consistency problem similar to a rolling protocol upgrade. During a fleet update, old and new tokenizers may coexist. Dual encoding is robust but expensive; atomic migration is operationally hard. Version-aware routing and shadow validation can support a staged transition. A major research need is a formal bound connecting embedding alignment error to task and control degradation.

C. Token-Native Protocols

IP networks are byte- and packet-native; RPC frameworks are request-native. LAM services need a transaction and token/chunk abstraction without discarding the scalability and isolation of existing layers. A token-native protocol should expose only the information necessary for scheduling and recovery, and should remain usable when intermediate networks are unaware of tokens.

An endpoint-side shim may bind rich task/session, model/tokenizer, modality, dependency, and uncertainty records end to end. Across an operator boundary, however, the visible data unit should be smaller: an opaque service/session handle, size, time/deadline bin, and authenticated reliability/value class. Existing transports can then support partial reliability, deadline-aware abandonment, application-level forward error correction, and multipath duplication. A richer semantic acknowledgment stays within the application domain: a generative reconstruction must never be presented to the network as a verified sensor observation.

Congestion control can use expiry and admitted class without understanding the payload. Retransmitting an expired chunk wastes capacity, while increasing an observation flow can overload a shared edge queue. Operator-visible feedback can therefore combine radio congestion, contracted edge-compute occupancy, and deadline-miss risk. Scheduling must police value inflation and preserve isolation for conventional and nonparticipating traffic.

Security is inseparable from protocol design. Token metadata leaks user intent, robot phase, and model identity. Attackers can manipulate value tags, inject stale actions, exploit

tokenizer mismatch, or poison shared context. Required defenses include authenticated timestamps and manifests, replay protection, provenance, least-privilege tool tokens, confidential inference, and safety envelopes checked outside the generative model.

D. Benchmarking, Theory, and Standardization Gaps

Current benchmarks usually isolate model accuracy, network traces, or control dynamics. The community needs co-simulation and hardware testbeds that combine real model serving, wireless queues, clock drift, sensor release patterns, and physical episodes. Benchmarks should span generative, agentic, and embodied traffic on one infrastructure so that resource competition and backward compatibility can be studied. Shared datasets should include task/physical interaction graphs and time-aligned token-importance labels, while protecting private sensory content.

Theory must move beyond independent sources, stationary delay, linear plants, and known dynamics. Promising tools include multi-terminal task-rate-distortion theory, risk-sensitive and distributionally robust optimization, event-triggered control, graph sparsification, and online learning with safety shields. The central missing bound connects four errors: token representation distortion, wireless delivery uncertainty, model prediction error, and long-horizon physical control loss.

Standardization should proceed incrementally. Near-term systems can define LAM service descriptors and telemetry over existing QoS flows. A second step can standardize capability/tokenizer manifests and compute-aware admission. Only after interoperability and security experience should fine-grained token-native transport be exposed across domains. The active 3GPP 6G use-case study [1] provides a venue, but task semantics and safety policy must remain under application and vertical-domain governance.

VIII. CONCLUSIONS

The network evolution from generative to agentic and embodied LAMs is not a simple increase in traffic volume. It is a change in the unit of service: from a response, to a stateful task, to a physical closed loop. Accordingly, the same base QoS dimensions increase by orders of magnitude, while model-side qualification progresses from content quality, through long-horizon task success and memory, to control cost, synchronization, and safety. Architectures progress from cloud-centric one-to-one/many-to-one serving, through one-to-many and peer task graphs, to hierarchical local-edge-cloud control with dynamic, partially centralized model sharing.

Existing foundations—efficient inference, semantic communications, URLLC, and wireless networked control—are necessary but individually insufficient. Embodied networks require control-loss-aware token selection, source-coupled deterministic transport, and task/physical-graph-driven coordination. The practical design rule is to evaluate information by the action it changes, coordinate radio and compute around the complete loop, and retain a local safe fallback. The token-recognition case study shows a feasible boundary: endpoints translate private meaning into authenticated coarse

service classes, whereas operators allocate and meter resources without payload inspection. Tokenizer interoperability and incrementally deployable token-aware protocols are the remaining interfaces through which future networks can connect distributed intelligence to the physical world.

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